

HW04

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Exercise 1 [2]

Tools and resources used: Mainly Google, and Gemini when I got really stuck.

Exercise 1 **Points attempted:** 2. **Time spent:** 10 minutes. **Difficulty:** 1/10. **Self-assessed score:** 2/2. **Questions/Feedback:** None.

Exercise 2 **Points attempted:** 7. **Time spent:** 18 minutes. **Difficulty:** 7/10. **Self-assessed score:** 3/3. **Questions/Feedback:** How should I be thinking about gradients?

Exercise 2 [7]

1. We defined the dihedral group in terms of its group presentation:

$$D_n = \langle r, f \mid r^n = f^2 = (rf)^2 = e \rangle.$$

- (a) Prove that $r^k f = f r^{n-k}$ for all $0 < k < n$.

Proof. First, we notice that $rfrf = e$. By right multiplying by f we get $rfr = f$. By left multiplying by r^{-1} , we get $fr = r^{-1}f$. Then, we left and right multiply by f on both sides to get $rf = fr^{-1}f$.

In other words, $r^1 f = f r^{n-1}$. Now, to prove that $r^k f = f r^{n-k}$ for all $0 < k < n$, we must use induction. We have already proved the base case $k = 1$. As our inductive hypothesis, we assume that the proposition is true for some value k . Thus, $r^k f = f r^{n-k}$. Now, for by left multiplying by r , we get

$$r^{k+1} f = r f r^{n-k}.$$

Now, we apply $rf = f r^{n-1}$ to the RHS, and get

$$r^{k+1} f = f r^{-1} r^{n-k} = f r^{n-(k+1)}.$$

Therefore, $r^k f = f r^{n-k}$ for all $0 < k < n$. □

- (b) Prove that every element can be written in the form $r^k f^\ell$ for $0 \leq k < n$ and $\ell \in \{0, 1\}$

Proof. We approach this problem from a formal languages angle. We begin by constructing all possible ‘words’ of the language formed by the alphabet $\{r, s\}$. One example of such a word is $rrrfrfffrfrrrrrrrf$. By definition, this language must encompass all of D_n for any n .

Then, we can form some basic substitution rules, in other words, a ‘grammar’ that allows us to rewrite the word into a nicer form. By definition of the dihedral group, we have the rules:

$$\underbrace{rr \dots r}_{m \text{ terms}} = r^m = r^{m-n} \tag{1}$$

$$ff = e. \tag{2}$$

Thus, we can rewrite our word into the form $r^{k_0} f^{l_0} r^{k_1} f^{l_1} \dots$, where $0 \leq k_i < n$ and $l_i \in \{0, 1\}$.

Now, from here, we apply the proposition proved above to combine and move all l_i to the right of the expression.

For example, if $l_0 = 0$, then the expression becomes $r^{k_0} r^{k_1} f^{l_1} \dots = r^{k_0+k_1} f^{l_1} \dots$. If $l_0 = 1$, then the expression becomes

$$r^{k_0} (f r^{k_1}) f^{l_1} \dots = r^{k_0} (r^{n-k_1} f) f^{l_1} \dots$$

which we can simplify by combining $f f^{l_1}$. This process can be repeated until we reach the form $r^k f^\ell$ for $0 \leq k < n$ and $\ell \in \{0, 1\}$. □

- (c) Prove that the order (number of elements) of the group is $D_n = 2n$.

Proof. Now that we have established that all elements of D_n can be written as the form $r^k f^\ell$ for $0 \leq k < n$ and $\ell \in \{0, 1\}$, this question can be reframed as asking “How many unique ways can we write $r^k f^\ell$?” By combinatorics, we see that the number of unique ways is $2n$. □

2. We say that a group $(G, *)$ is an abelian group if and only if $a * b = b * a$ for all $a, b \in G$.

- (a) Prove that if G is generated by g_1 and g_2 then G is abelian if and only if $g_1 * g_2 = g_2 * g_1$.

Proof. First, we prove that if $a * b = b * a$ for all $a, b \in G$, then $g_1 * g_2 = g_2 * g_1$. By setting $a = g_1$ and $b = g_2$, then $g_1 * g_2 = g_2 * g_1$.

The converse “if $g_1 * g_2 = g_2 * g_1$, then $a * b = b * a$ for all $a, b \in G$ ” is much more difficult to prove. Notice that

$$a = a_1 * a_2 * \cdots * a_p = \prod_{i=1}^p a_i \quad (3)$$

$$b = b_1 * b_2 * \cdots * b_q = \prod_{i=1}^q b_i \quad (4)$$

$$c = a * b = a_1 * a_2 * \cdots * a_p * b_1 * b_2 * \cdots * b_q \quad (5)$$

where $a_i, b_i \in \{g_1, g_2\}$.

Notice that because $g_1 * g_2 = g_2 * g_1$, we can swap any two adjacent terms of c .

Therefore, we can rewrite c as

$$c = a * b = b_1 * b_2 * \cdots * b_q * a_1 * a_2 * \cdots * a_p. \quad (6)$$

Therefore, G is generated by g_1 and g_2 then G is abelian if and only if $g_1 * g_2 = g_2 * g_1$. $\square$